

Scale Economies and Welfare Effects of Large-Scale Landlords under Heterogeneous Renter Preferences

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IO Student Lunch



Heterogeneous Preference and Gentrification

- Improvements in local attributes (e.g., environmental cleanup, crime reduction) change housing demand and induce new sorting equilibria
- Higher-valuation households move in, raising housing prices [Autor et al., 2019, Greenstone and Gallagher, 2008]

This paper: supplier-driven attribute changes

- What if certain housing attributes are tied to specific landlord types?
- Institutional landlords may have lower costs of providing location convenience:
 - Better access to well-located properties (e.g., near campus)
 - Scale advantages in acquisition, marketing, and portfolio management
- How does institutional ownership affect equilibrium rents and renter welfare?

Literature Review: Institutional Real Estate Investors

Institutional investors have expanded rapidly in both single-family and multifamily rental markets since the Great Financial Crisis.

- Coven et al. [2025], Bogin et al. [2022]: Institutional purchases crowd out marginal homebuyers, reducing homeownership rates.
- Coven et al. [2025]: Institutional entry increases rental supply and lowers average rents in equilibrium.
- Barbieri et al. [2024]: Institutional landlords has market power in markets where they hold significant local market share. They are also more efficient in vacancy management.
- Waldinger [2024], Amborski [2020]: Institutional landlords benefit from economies of scale in property management, maintenance, and leasing.
- Amborski [2020], Mills et al. [2019]: Institutional ownership increases rent dispersion and strategic pricing due to portfolio optimization.



- Extend the framework beyond student housing to general rental markets
 - The definition of “location convenience” may differ across contexts
 - Key heterogeneous attribute: distance to city centers
- Supply-side revision: landlords jointly choose portfolio composition and prices
- No first-mover advantage. Large landlords and individual landlords only differ in portfolio holding cost.

Altos Rental Listings Data

- Weekly listing-level panel of rental units
- Observed outcomes:
 - Listed monthly rent
 - Listing appearance and disappearance
- Unit and location characteristics:
 - Address and geolocation (street, ZIP)
 - Property attributes (beds, baths, square footage, parking, property type)

CoreLogic Ownership Data

- Matched to Altos listings using property address
- Identifies ownership type (institutional vs. mom-and-pop)
- Owner names enable the construction of portfolio

Motivation

Renters differ in willingness to pay for location convenience.

Two renter types

- Type H : High commute disutility (large $|\gamma^H|$)
- Type L : Low commute disutility (small $|\gamma^L|$)

Let the share of type H renters in market t be π_t .

Utility for renter i of type $\tau \in \{H, L\}$

$$u_{ij\tau t} = -\alpha p_{jt} + \beta' X_{jt} - \gamma^\tau \text{Distance to City Center}_{jt} + \xi_{jt} + \varepsilon_{ij\tau t}$$

Observable property attributes from the data

- X_{jt} : observable characteristics of property j in period t

Type-specific mean utility

$$\delta_{jt}^{\tau} = -\alpha p_{jt} + \beta' X_{jt} - \gamma^{\tau} \text{Distance to City Center}_{jt} + \xi_{jt}$$

Type-specific lease probability (logit)

$$q_{jt}^{\tau} = \frac{\exp(\delta_{jt}^{\tau})}{1 + \sum_{k \in \mathcal{J}_t} \exp(\delta_{kt}^{\tau})}$$

Observed lease probability (mixture)

$$q_{jt} = \pi_t q_{jt}^H + (1 - \pi_t) q_{jt}^L$$



Institutional expansion affects welfare through

- Institutional landlords have a lower cost of providing location convenience (i.e., constructing a low Distance to City Center _{jt} portfolio) due to scale and better access to well-located properties.
- They may also increase total rental supply.

Key prediction

If institutional expansion raises rents but reduces commute time, Type L renters (who are more willing to commute in exchange for lower rent) may experience net welfare losses.

- Compute Distance to City Center $_{jt}$ from geolocation:

$$\text{Distance to City Center}_{jt} = f(\text{lat/lon of listing } j, \text{ job center / campus})$$

- Portfolio-level convenience:

$$\overline{\text{Distance to City Center}}_t = \frac{1}{|\mathcal{J}_t|} \sum_{j \in \mathcal{J}_t} \text{Distance to City Center}_{jt}$$

Institutional ownership (CoreLogic)

- Indicator $\mathbf{1}\{j \in I\}$
- Ownership portfolio mapping

Portfolio and pricing decision

- Landlord chooses price and portfolio:

$$\{p_{jt}, a_{jt}\}_{j \in \mathcal{J}}$$

- $a_{jt} \in \{0, 1\}$ indicates whether the landlord invests in / holds a unit at location j in period t :
 - $a_{jt} = 1$: include location j in the portfolio
 - $a_{jt} = 0$: do not invest in location j
- Holding a given portfolio incurs a fixed cost:

$$\kappa(\{a_{jt}\})$$

which depends on the average commute time of the portfolio.

- Institutional landlords face lower fixed costs for operating convenient portfolios.



Profit

$$\pi_l = \sum_j a_{jt} \left[(p_{jt} - c_j) q_{jt}(p_{jt}, x_j, \text{Distance to City Center}_{jt}) \right] - \kappa(\{a_{jt}\})$$

Demand

- q_{jt} is the probability that unit j leases in period t
- Depends on:

$$q_{jt} = q(p_{jt}, x_j, \text{Distance to City Center}_{jt})$$

- Higher price $\Rightarrow$ lower lease probability
- Better location (shorter commute time) $\Rightarrow$ higher lease probability

Supply-side tradeoff

- Portfolio choice affects demand through commute convenience
- Landlord trades off:
 - Higher revenue from convenient locations
 - Higher fixed cost of operating such a portfolio
- Institutional landlords have lower fixed cost $\kappa(\cdot)$:
 - More likely to operate convenient portfolios

- **Counterfactual: Welfare impact of landlord scale composition**
 - Goal: measure the welfare change for type L renters (who value scale-related attributes less) when the market share of large-scale landlords increases
 - Compute equilibrium rents, allocations, and renter welfare under alternative landlord scale compositions
- **Policy experiment to shift landlord scale composition**
 - Landlord market share is an equilibrium outcome
 - Identify policies that exogenously affect the relative prevalence of large-scale landlords (e.g., regulation or tax treatment)
- Construct and validate Distance to City Center_{*jt*}
- Construct a proxy for π_t (share of high-type renters in market t)
 - Use renter demographic composition (e.g., student share, income, occupation)
- Reduced-form evidence: large-scale landlords operate relatively more in high-access locations

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